DISCUSSION OF WINDING PROCESS FUNDAMENTALS

Understanding winding, roll defects (e.g. tin-canning), and winding-related web quality (e.g. bagginess) begins with understanding how the web, winder design, and process conditions combine to create the stress and strains of the wound roll.

To understand the stresses and strains within a wound roll we must understand four things:

1. What determines the web tension as it makes first contact with the winding roll?
2. How is the tension controlled from the first to final lap of the winding roll?
3. How do dimensional changes within a roll (e.g. compression) change roll strains and stresses?
4. How do transverse direction variations in equipment or web properties affect roll structure?

PART 1: WINDING CONTROL: CREATING WOUND-ON TENSION (WOT)

The tension of the web as it first makes contact with the winding roll (called the wound-on tension or WOT) is a strong function of two variables: center torque and surface nip load.

Understanding how applied center torque (in force x radius) becomes tension is easy. Torques are a force times a distance. Reversing this relationship, we can find the force of winding tension will be center torque divided by roll radius.

Understanding how surface nip load creates tension is less straightforward. Nipping the web as it enters the winding roll elongates the web much like a rolling pin on cookie dough. Just as cookie dough must slip on the baker's sheet to stretch, so too must the web slip relative to the layer below it in the winding roll. This dependence on slippage makes the relationship of nip load to tension friction dependent. For low to moderate nip loads (<10PLI or <2000 N/m), a good estimate of tension created by nipping (a.k.a. nip-induced tension or NIT) is found by multiplying the nip load (in force/width) by the coefficient of friction of the product's side A to side B. For a product with a coefficient of friction of 0.25, nipping with 4 PLI (700N/m) is like turning up the tension 1PLI (175 N/m).

The total wound-on tension is the sum of tension from center torque and nip-induced tension.

M_{CW} = Center Wind Torque (in-lbs or N-m)

r = Radius (in or m)

W = Width (in or m)

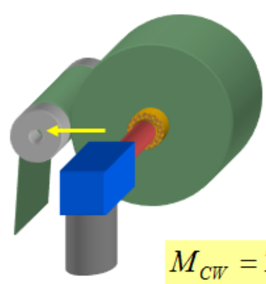
T_{WOT} = Wound On Tension (lb/in or N/m)

μ_K = Web Coefficient of Friction

N = Winding Nip Load (lb/in or N/m)

T_{WH} = Web Handling Tension (lb/in or N/m)

$$T_{NCW,WOT} \approx \frac{M_{CW}}{rw} + \mu_K N$$



$$M_{CW} = T_{WH} wr$$

The above relationships of center torque and nip load apply to surface winders and combination center-surface winders with a couple additional considerations. For surface winders, there is no center torque, so that term is eliminated. Also, some surface winders may not be completely isolated from upstream web handling tension. Adhesion, mechanical entanglement, or gravity may change how the web slips on the surface drum, increasing the WOT above what nip load alone can account for.

PART 2: WOUND-ON TENSION VS. RADIUS AND SPEED

The wound-on tension (WOT) will normally change for each layer of a winding roll. If applied torque and nip load are constant, the torque contribution to WOT will reduce with growing roll radius. Many winding machines include options to program a changing tension or nip set point as a function of roll radius or speed.

Winding torques and tensions may be controlled to vary with roll radius. In constant tension winding, the center torque is increase proportional to roll radius. In constant torque winding, torque is held constant and winding tension will drop inversely with roll radius. Many wound rolls are wound with a tension vs. radius controlled somewhere between the constant tension and constant torque conditions. The profile of tension vs. radius is usually called winding taper tension or tension profile.

There are many logical reasons to change winding torque or nip load during the winding process. Some of these include:

- All the layers of a winding roll must transmit the applied center torque through the layer-to-layer frictional contact without slippage. The layer near the core are especially challenged in this ability to transmit torque since they have less area per layer to develop friction and they are at a mechanical disadvantage to the outer layers of a roll. Therefore, center wound and unwound roll need more pressure in near core layers. To achieve this near core tightness, many winding processes use higher values for winding tension and nip load when winding near core layers.
- In winding smooth, non-porous products, an air layer will winding into the roll without sufficient pressure to compress or squeegee the entrained air. If excess air is allowed in a winding roll, the roll structure will loosen over time as the air escapes the wound roll, leading to soft roll defect (TD Buckling, sag, roll handling or unwind telescoping). A winding nip roller is an effective tool to greatly reduce air captured in a winding roll. The air exclusion of a winding nip is directly a function of speed and inversely a function of effective radius (which is a factor of both nip roller radius and winding roll radius). To winding a roll with constant entrained air per layer (with a goal of controlling entrained air to less than the surface roughness of the product), nip load would need to be profiled to both speed and roll radius.
- The nip-induced tension (NIT) may be a function of roll radius with more NIT when roll radius is small.
- Entrained air will be a function of speed. Many winders, including most slitter rewinders and all zero-speed core transfer winders must start and end the winding roll at zero speed. Should winding torque and nip load be controlled as a function of speed?

Clearly, if winding torque and nip load are held constant they will not create a constant roll structure due these speed and radius effects. However, constant roll structure is not necessarily the goal of winding. Just as a skyscraper will have different demands for the building structure of layer near the bottom vs. top, so too will winding rolls have different structure demands from core to final layer.

Based on my experience with many film products, the best and simplest approach to creating a robust roll structure is to eliminate any speed functions on both tension from center torque and nip, to run nip load at a constant value that prevents excessive air at the maximum radius and speed conditions, and to taper tension vs. radius with final tension between 20 and 60% of starting tension on a linear tension or torque vs. radius curve. This approach creates a roll with near core layers that are tighter and have less entrained air and outer roll layers, while also reducing the torque transmission requirement in winding the final layers, but having sufficient nip load to avoid shifted layers of air lubrication.

PART 3: FINAL ROLL STRUCTURE: UNDERSTANDING INTERNAL STRESSES AND STRAIN

Most wound rolls are held together with the friction. The tensioned (elongated) layers create pressure pushing towards the core, much like the stretched rubber of an inflated balloon pushes against the balloon's internal air pressure. Tension is proportional to the web strain (defined as the percent change in length from the web's untensioned length). The pressure of one layer equals web tension divided by roll radius (P=T/R, with tension in units of force per width).

As a roll builds, each layer can add to the tightness of the roll. One layer wound on a core may create a low pressure (less than 1 psi or 7 kPa), but a wound roll may have thousands of layers and potential interlayer pressure greater than 1000 times the pressure from one layer. Where 1 psi (7kPa) may not be enough pressure to hold a roll together against forces of winding, handling, and unwinding, 1000 psi (7MPa) is certain excessive and cause unneeded defects, such as impressions, core crushing, or blocking of layers within the roll.

In most winding, the adding of 1000 layers does not increase the roll tightness by 1000 times. In winding a soft film, like polyethylene, you might get close. In winding papers, you won't' come close. The difference in winding these two materials is due to strain and strain losses (Remember 'strain is the secret to web handling?').

Low modulus films will have more strain for a given tension and require more compression or radial shifting to lose their in-roll tension. Since the stack of polyethylene is relative stiff compared to the machine direction modulus, it won't compress enough to big strain losses. This combination makes polyethylene very easy to wind too tight.

Papers are stiff and will have low strains for a given tension. Papers will also be relative soft in the stack direction and can have large strain losses from compression. This combination makes it difficult to winding an extremely tight roll of paper.

The winding torque and nip load determine the initial tension for a layer as it enters the winding roll, but the stress and strain within the roll of any given layer may be much lower than it initial wound-on tension, often approaching zero tension (circumferential stress in the roll) or even going below zero into circumferential compression).

The loss of circumferential tension is due to radial shifting or compression within the winding roll. Layer typically shift to a lower radial position due to core compression, compression of inner roll layers from the added pressure of outer roll layers, compression of surface roughness features of the product, density increases in the product, air escaping the roll, and lateral flow of any layers (such as adhesive layers in sticky tape products). The compressibility of the winding roll is characterized in stack compression testing and is a critical product mechanic property required for advanced understanding of winding any product.

The stack modulus (considered the radial modulus of a winding roll) is different from tensile or Young's modulus of elasticity (which is the circumferential modulus in the winding roll) in two ways. First, stack modulus is not a constant, increasing under higher pressures. Second, stack modulus is often much lower than tensile modulus (except in highly compressible soft film stacks).

Wound-on tension, roll geometry, and stack modulus are all needed to model internal roll stresses and strains as function of radial position. In most cases, interlayer pressure will be highest at or near the core and decreasing to zero at the outside of the roll. Internal roll tensions will vary greatly by product and winding condition, but will be low through most of the roll with the greatest variations at or near the core depending on the magnitude of compression vs. the initial strain of wound-on tension.

One of my constant problem solving tips of web handling and winding is that ‘secret to web handling is strain.’ The loss of in-roll tension and pressure from compression occurs with what seem like minor radial changes. However, a 0.2% strain is the same magnitude as a 7 mils (0.1mm) shift in a layer at a radius of 4-inches (200mm). For materials that are stiff in the machine direction, but relatively soft in the thickness direction, radial compression of web, surface roughness, and entrained air will greatly reduce the tensions and pressures within a wound roll. Many layers of a wound roll is have lost over 90% of their wound-on tension, with near core layers commonly with zero tension or even pushed in to circumferential compression.

Even when winding is finished, the roll will likely continue to have changes in internal roll stresses and strains. If rolls were purely elastic, they would remain in the same stress and strain condition until unwinding; however, several factors can make a wound roll structure change over time.

Entrained air and its escape is one of the most common causes for roll structure changes over time. Excess air wound into a roll will slowly escape over times of hours, days, or weeks. As the air escapes, layers will fall towards the core and lose their machine direction tension. As the web loses MD tension it will attempt to regain Poisson’s effect width losses. In some materials, this width recovery will appear as MD buckles in the wound roll (a.k.a. tin-canning defects).

Other potential dynamic effects on winding include: visco-elastic recovery of pre-winding tension, creep of visco-elastic materials (especially adhesive coated products), and any change than will change the length, thickness, or width of layers within the roll, including hygro and thermal expansion.

PART 4: ROLL TRANSVERSE DIRECTION (TD) VARIATIONS

The above discussion covers machine direction (including radial position) and time dependent variations in rolls, lastly we need to address transverse direction (TD) variations, especially since many roll and web defects have TD variations.

In accessing TD variations within a roll, we imagine dividing a winding roll in narrow lanes much like dividing a wide roll into narrow slit rolls.

Different TD lanes will have different average thickness and will build to differing diameters. Lane-to-lane of a wide winding roll or roll-to-roll diameter variations of slit rolls create surface speed variations. When the web sees increasing or decreasing speed the web strain and stress will follow. Lower than average diameter lanes will have below average tension; above average diameters will have above average tension. Nipping will vary TD from both diameter variations and uneven nip loading. Lower than average diameter lanes will have below average nip load; above average diameters will have above average nip load.

Many factors in winding are non-linear, so small changes in thickness can create 10x to 50x higher stresses and strains in thick vs. thin lanes. The biggest non-linear factor in winding is the stack modulus. A stack of most materials loaded with minimal pressure is initially soft and easy to compress. However, as a stack is pressed into a tighter stack it becomes more difficult to compress. Then, under extremely high pressure, the stack will have squeezed out all the voids of entrapped air, porous structures, or surface roughness and you will reach a compressibility of the bulk material. This transition from spongy to full compressed stack will be non-linear when viewed on a pressure vs. strain plot.

In winding a roll with TD thickness variations, the thick lanes will be under higher tension and stresses for two reasons. First, the larger diameter lanes formed from thicker than average material will have high wound-on tension since the above average diameter will steal tension from the other lanes of the roll and have an above average nip load and nip-induced tension. Second, the higher tension from torque and nipping will form a stiffer stack in the thick lanes of winding. Due to higher starting tension and less compression to relieve the wound-on tension, the thick lanes of a wound roll will have non-linearly higher stresses and strain than average or below average thickness lanes.

These variations would create less problem is the web reacted elastically. If you look at the strain differentials within a roll, they don’t appear to exceed the material’s yield point. If the same strains occurred while wrapping a roller, the web would pass by undamaged. However, since the web is held under these stress and strain conditions for the long times of storage and transport, the web’s response is not elastic.

In materials science, creep is the tendency of a solid material to move slowly or deform permanently under the influence of stresses. If different lanes of a web were be strained at 0.5 percent vs. 0.2 percent (or zero percent in loose lanes) for a short time period, such as when wrapping a concave roller, and the yield strain was 2 percent, there would be no permanent damage to the web and the lay flat or bagginess would be unchanged. However, if the web was held under the same strain differential conditions for long periods of time, such as within a wound roll, a portion of the strain would become permanent. For example, all lanes might retain 10% of their long-term train, setting in a permanent strain of 0.05 and 0.02 percent.

These may seem like small dimensional differences, but the length variations of 200 and 500 parts per million (200 micro-in per inch or 200 microns per meter) are enough to create an obvious baggy lane. For a stretch material, this level of bagginess would be pulled out at modest tensions. For a high modulus material, such as bond paper or polyester, it would require 0.25-0.75 PLI/mil to pull out this bagginess (1.7-5.1 MPa). Surprisingly, it would take more tension to pull out bagginess if there are only a few baggy lanes and more tension to pull out a web with a great percent its width that it baggy. This makes sense we you realize that bagginess goes away by stretch the short lanes to a length equal to the long lanes. Therefore, more percent width of bagginess means there is less width of short material needed to be pulled to the baggy lanes length.

There are other transverse direction factors that may create cross-roll stress and strain variations, including:

- Baggy input web
- Uneven nipping left-right
- Uneven nipping due to core or nip roller deflection
- Uneven nipping due to core diameter variations
- Uneven tensioning from roller or core misalignment
- Wrinkles from a poor core start (quite common with flying knife auto roll transfer systems)
- Wrinkles wound into the roll, often in the center 80% of the roll width, but not near the edges.
- Poor slit edge quality on one or both edges
- Uneven bulging or shrinkage of the winding core.

The result of TD variation in wound rolls is defects that may form in specific TD positions (e.g. near one edge, in the center of the roll's width, aligned to a thick or thin lanes, between thick lanes.

The combination of TD variations and natural changes in roll structure from core to outside layers may also create defects in specific combination of TD and radial positions.

SUMMARY

Solving wound roll defects is complicated. However, by understanding these four fundamental areas of winding [1) wound-on tension, 2) profiled control of tension and nip load, 3) understanding the transitions to wound-in stresses, and 4) mapping TD variations onto your wound rolls] you have the tools to apply our best knowledge of the winding process to improve your product quality and yields.

HOW THE OPEN LOOP TORQUE CONTROL SETS WINDING TENSION

There are many winding machines that do not use closed-loop tension control in the winding process. Most of these machines request input of starting and taper functions of winding tension and calculate the complex function desired tension versus roll diameter or winding speed. These programs convert the calculated tension into motor torque applied at the torque to create the outer roll tension vector. Hopefully these open-loop torque control winding systems take care to consider torque losses, such as rolling resistance or inertial losses during acceleration and deceleration.

For rolling resistance, some systems include a setup feature to perform individual winding spindle tests to measure and record rolling resistance as a function of speed and control winding torque to a value above the baseline of rolling resistance. This is somewhat common, but rarely do these systems look to measure rolling resistance of full diameter or maximum weight rolls, instead assuming rolling resistance is independent of roll weight or winding nip load.

All this capability is great and quite advanced compared to many slitters where tension is controlled by friction torque systems, but it requires a great deal of trust that the slitter control engineers understand what motor control parameters create a given torque output over a span of speeds and loads. This open loop system has no feedback method to indicate process changes from equipment wear, failure, calculation error, except in the quality of wound rolls. It would be great to be able to verify the system performance with a tension measuring roller. A feature of many slitter rewinders, including this one, is a short web path from slitting to winding, ensuring good winding roll sidewall, but leaving no consideration for a winding tension measuring roller.

My preference would be to at least have an optional web path from slitting to winding that would include tension measurement. This option would allow for verification trials confirming the open loop torque control system is creating the desired tension and the combination of miscalculations, bearing failure, roll weight, nip load, and inertial factors do not cause significant errors or variations.

THE CASE FOR SHORTER (AND LONGER) ACCELERATION TIMES

I recently visited a slitter-rewinding that was set with an acceleration time of 180s (3 minutes) to reach 1000 fpm (300 mpm). I recommended changing to a faster acceleration time for several reasons:

- Many thin products, including the one that was running on this slitter, wrinkle much more at low speed. At higher speeds, air lubrication reduces the web to roller traction, reducing the ability of a roller to shift or gather the web or hold the web in a buckled, wrinkled shape.
- I was surprised to see the acceleration set in time (seconds). In my experience, most converting machines have acceleration set in units of velocity per time (mpm/s or fpm/s), not simply time. For example, limiting acceleration to 10 mpm/s would reach 100 mpm in 10s, 300mpm in 30s, and 800mpm in 80s. Setting an acceleration time creates a slower acceleration rate for slowing speed set points. I would prefer to see a single acceleration speed independent of target speed.
- This particular slitter was also set up to have a two stage acceleration, first accelerating to and running at a low speed, then when the slitting winding process looks good, accelerating to the higher target speed. I can understand this approach for developing a set of winding conditions, when you might see in the first few wraps that wrinkling or shifting layers is a problem and want to stop, but the ultimate goal for slitter winding should be a rapid acceleration to line speed without an intermediate speed. The extra time at low speed and inertia upsets of speed changes will create undesired tension variations and roll defects.
- The target accelerations rate should be much faster than 300mpm/180s. Many slitter-rewinders are designed to acceleration at 100fpm/s (30mpm/s) – 18x faster than this slitter was set to run reaching 1000 fpm in 180s. I recommend at least 30 fpm/s (10 mpm/s) or reducing acceleration time to 30s to reach 1000 fpm (300mpm). Ultimately, you may be able to reach 1000 fpm (300mpm) in 10s, greatly reducing the time and waste at low speeds.

The case against rapid acceleration.

- Many advocates of rapid acceleration do so in the name of machine output. Obviously, any time spent at lower speed does not process as much material as higher speeds. In the case of extremely short slit rolls one fully automated slitter rewinders, then there is a strong correlation between accelerations times and output. However, for slitters running long roll lengths, there is extremely small output benefit of reducing the acceleration times.
- A machine that runs larger diameter heavy rolls and does not have inertia compensation (such as many torque controlled unwinds and rewinds) will see dramatic tension variations during rapid acceleration.
- The added torque of rapid acceleration can put unnecessary loads on coupling, driven shaft, belts, chucks, shafts, and other components, leading to early failure and associated downtime and costs. Also, idling and driven roller are more likely to slip during rapid acceleration, leading to roller wear or product abrasion (generating surface contamination and scratching).
- Rapid acceleration will always be a challenge to closed-loop tension control systems, creating tension transients high or low, especially without optimizing the PID or other control algorithm.
- Tension and speed transients during acceleration are commonly associated with registration losses in printing and die cutting and coating variations. Wrinkling can also occur in the changes from high to low tension as the Poisson’s effect alters the web width from narrow to wide.

WAR ON BAGGINESS

Bagginess is an area that many paper, film, and foil converters struggle with. Due to many factors - including the lack of good bagginess quantified measurements, need for feedback from roll unwinding measurements/observations (usually at customer processing), and complexity of process and material properties that create bagginess forms – bagginess continues to plague many web producers as an uncontrolled variation in product quality.

I have worked closely with Dr. Kevin Cole of Optimation Technology Inc. (formerly the winding and web handling corporate expert at Kodak) on understanding bagginess in films and foils. It is truly a highly complicated defect, but one that we now have many tools in place to address.

Here is combination of product and process variables needed for a full bagginess reduction program:

- On-line thickness profile measurement confirmed with off-line thickness measurement to understand the profile variation in the time span of winding a single roll.
- Understanding of material properties, including coefficient of friction, surface roughness, stack modulus pressure-compression behavior, and visco-elastic creep function (on a days or month scale, though this can be estimated from fitting bagginess measurements onto model predictions).
- Three-dimensional winding model to predict interaction of roll geometry, material properties, winding process variables (tension, nip, speed, taper), and transverse direction thickness profile.
- The 3D winding model predicts machine direction strains at all radial and transverse positions in the wound roll, then maps how much of these strains will become non-elastic permanent strains in the product upon unwinding, where transverse direction permanent strain variations will be observed as bagginess.
- Some bagginess trends that are common in films and predicted by these models include:
  1) Loose baggy lanes will align to thickness variations 'up' features. Quantifying thickness variations with a simple range (minimum to maximum) statistic will be a somewhat poor predictor of bagginess magnitude. Peak to average statistics, emphasizing thick bands over thin bands, will be a better bagginess predictor.
  2) Bagginess will be greatest in a roll's outer wraps.
  3) If compressible cores are used, the high pressure thick lanes can indent the core, creating an opposite bagginess trend of tight lanes corresponding to thickness 'up/thick' features.
  4) Winding with more tension and nip load will have high average strain and strain variations within a roll, creating more permanent strain variations and bagginess.
- Bagginess measurement is the final key tool and one that remains either difficult or expensive. Even qualitative bagginess can be quite time consuming. There are many options to measure bagginess, but the best systems will quantify bagginess in terms of the length change from baggy lanes to non-baggy lanes, much like a thickness gauge documents crossweb thickness profile. In units of percent length differential, these measurements can then be mapped on to the 3D winding model predictions and confirm the combination of winding, film, and thickness variations required to lower bagginess to a desire level (i.e. pulled out with a given tension).

Reducing bagginess is a challenging program, but begins with quantifying bagginess (to measure when you have improved), reducing thickness variations, and minimizing winding stresses and strains.

FUNCTIONS OF A NIP ROLLER AT WINDING (UNFINISHED)

A winding nip roller has five effects, most beneficial.

1) The winding nip adds tension, creating more wound-on tension than center torque alone.
2) With proper wrinkle prevention or spreading into the nip roller, the nip will lay the web down wrinkle-free on to the winding roll, preventing the roll variations (e.g. TD diameter variations) from gathering or shifting a web so severely it wrinkles.
3) With proper guiding or a short web path from slitting to winding, the nip will lay the web down with good alignment, preventing the roll variations (e.g. TD diameter variations) or uneven escape of air to shift the web.
4) The winding nip reduces the air that enters the winding roll, preventing roll stress and structural change from excess air escaping over time.
5) A winding nip may promote better cylindricity, especially in products with low stack modulus.

Things that go wrong with nips:

1) Deflection
2) Wrinkling
3) Damage.